

Math 4441: Probability and Statistics

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Chapter 4
Conditional Distributions, Conditional Expectations and Independence
(Part 1 – Discrete Conditional Distributions)
Lectures 20 – 21

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References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Introduction

- A conditional distribution is a complete probability model, and it follows the axioms of probability
- It is called conditional distribution because it is conditioned on the availability of some partial information
- One of the most important topics in probability, because
 - We are often interested in probabilities or expectations with some partial information available
 - It is often useful to assume the availability of some partial information

Conditional distributions extend the conditional probability explained in Chapter 1 for Random variables

Conditional probabilities

Conditional probability

$$P[A|B] = \frac{P[AB]}{P[B]}, P[B] > 0$$

Product Rule or Law of Multiplication

$$P[AB] = P[A|B]P[B] = P[B|A]P[A]$$

Sum Rule or Law of Total Probability

$$P[A] = \sum_{i=1}^n P[A|B_i]P[B_i]$$

Bayes Theorem

$$P[B_i|A] = \frac{P[A|B_i]P[B_i]}{\sum_{i=1}^n P[A|B_i]P[B_i]}$$

Independence

$$P[AB] = P[A]P[B]$$

Conditional Distribution

1. Conditioned on an event

Experiment: Roll a fair six-sided die

$$S_X = \{1, \dots, 6\}$$

$$A \triangleq \{\text{event that the outcome is an even number}\}$$

$$P[A] = \frac{1}{2}$$

$$P_{X|A}(x) = \frac{P_X(x)}{P[A]}, P[A] > 0$$

x	1	2	3	4	5	6
$P_X(x)$						

$$A = \{2, 4, 6\}$$

$x \cap A$	2	4	6
$P_{X A}(x)$			

$$P_X(x) = \frac{1}{6}, \quad x = 1, \dots, 6$$

$$P_{X|A}(x) = \begin{cases} \frac{1}{3}, & x = 2,4,6 \\ 0, & \text{otherwise} \end{cases}$$



2. Conditioned on another Random Variables

$$P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$P_{Y|X}(y|x) = \frac{P_{XY}(x,y)}{P_X(x)}$$

$P_{XY}(x,y)$	$y = 1$	$y = 2$	$P_X(x)$
$x = 1$	0.4	0.3	
$x = 2$	0.1	0.2	
$P_Y(y)$			

$P_{X Y}(x y)$	$y = 1$	$y = 2$
$x = 1$		
$x = 2$		

$P_{Y X}(y x)$	$y = 1$	$y = 2$
$x = 1$		
$x = 2$		

$$P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)}$$

Product Rule or Law of Multiplication

$$P_{XY}(x, y) = P_{X|Y}(x|y)P_Y(y)$$

$$P_{XY}(x, y) = P_{Y|X}(y|x)P_X(x)$$

Example 6.11. Professor May B. Right often has her facts wrong, and answers each of her students questions incorrectly with probability $\frac{1}{4}$, independently of other questions. In each lecture, May is asked 0, 1, and 2 questions with equal probability of $\frac{1}{3}$. Let X and Y represent the number of questions May is asked in a lecture, and the number of questions she answers wrong in that lecture, respectively.

1. Find the joint PMF of X and Y , $P_{XY}(x,y)$.
2. Find the probability of at least one wrong answer.

$$X \sim$$

$$P_X(x) =$$

$$Y|X = x \sim$$

$$P_{Y|X}(y|x) =$$

$$P_{XY}(x,y) = P_{Y|X}(y|x)P_Y(y)$$

$P_{Y X}(y x)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$				
$x = 1$				
$x = 2$				

$P_{XY}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$				
$x = 1$				
$x = 2$				
$P_Y(y)$				

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$	$\frac{16}{48}$	0	0	$\frac{16}{48}$
$x = 1$	$\frac{12}{48}$	$\frac{4}{48}$	0	$\frac{16}{48}$
$x = 2$	$\frac{9}{48}$	$\frac{6}{48}$	$\frac{1}{48}$	$\frac{16}{48}$
$P_Y(y)$	$\frac{37}{48}$	$\frac{10}{48}$	$\frac{1}{48}$	1.0

$$\begin{aligned}
 P[A] &= P_{XY}(1, 1) + P_{XY}(1, 2) + P_{XY}(2, 2) \\
 &= \frac{4}{48} + \frac{6}{48} + \frac{1}{48} \\
 &= \frac{11}{48}.
 \end{aligned}$$

Example 6.12. Consider four independent rolls of a 6-sided die. Let X be the number of 1's, and Y be the number of 2's obtained. What is the joint PMF of X and Y ?

Solution. We use the multiplication rule to calculate the joint PMF $P_{XY}(x, y)$, which is the product of the following quantities.

1. The conditional PMF of X , $P_{X|Y}(x|y)$, conditioned on the value of Y , $Y = y$.
2. The marginal PMF of Y , $P_Y(y)$.

$$Y \sim$$

$$P_Y(y) =$$

$$X|Y = y \sim$$

$$P_{Y|X}(y|x) =$$

$$P_{XY}(x, y) = P_{X|Y}(x|y)P_Y(y) = \binom{4-y}{x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{4-y-x} \binom{4}{y} \left(\frac{1}{6}\right)^y \left(\frac{5}{6}\right)^{4-y}$$

Example 6.13. If X_1 and X_2 are two independent binomial random variables with respective parameters (n_1, p) and (n_2, p) , calculate the conditional PMF of X_1 , given that $X_1 + X_2 = m$.

$$\begin{aligned} P[X_1 = k | X_1 + X_2 = m] &= \frac{P[X_1 = k, X_1 + X_2 = m]}{P[X_1 + X_2 = m]} \\ &= \frac{P[X_1 = k, X_2 = m - X_1]}{P[X_1 + X_2 = m]} \\ &= \frac{P[X_1 = k, X_2 = m - k]}{P[X_1 + X_2 = m]} \end{aligned}$$

Since X_1 has the Binomial distribution with parameters (n_1, p) , therefore,

$$P[X_1 = k] = \binom{n_1}{k} p^k q^{n_1-k}$$

Similarly, for X_2 , we have

$$p[X_2 = m - k] = \binom{n_2}{m - k} p^{m-k} q^{n_2-m+k}$$

$$P[X_1 + X_2 = m] = \binom{n_1 + n_2}{m} p^m q^{n_1+n_2-m}.$$

$$\begin{aligned}
 P[X_1 = k | X_1 + X_2 = m] &= \frac{\binom{n_1}{k} p^k q^{n_1-k} \binom{n_2}{m-k} p^{m-k} q^{n_2-m+k}}{\binom{n_1+n_2}{m} p^m q^{n_1+n_2-m}} \\
 &= \frac{\binom{n_1}{k} \binom{n_2}{m-k}}{\binom{n_1+n_2}{m}}.
 \end{aligned}$$

Sum Rule or Law of Total Probability

$$\begin{aligned} P_X(x) &= \sum_y P_{XY}(x, y) \\ &= \sum_y P_{X|Y}(x|y) P_Y(y) \end{aligned}$$

$P_{X Y}(x y)$	$y = 1$	$y = 2$	$P_X(x)$
$x = 1$	$\frac{4}{5}$	$\frac{3}{5}$	$\frac{7}{10}$
$x = 2$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{10}$
$P_Y(y)$	$\frac{5}{10}$	$\frac{5}{10}$	1.0

$$P_X(x) = \sum_y P_{XY}(x|y)P_Y(y)$$

$$P_X(1) = P_{X|Y}(1|1)P_Y(1) + P_{X|Y}(1|2)P_Y(2)$$

$$P_X(2) = P_{X|Y}(2|1)P_Y(1) + P_{X|Y}(2|2)P_Y(2)$$

Conditional Expectation

$$E[X] = \sum_x x P_X(x)$$

$$E[X|A] = \sum_x x P_{X|A}(x).$$

$P_{X Y}(x y)$	$y = 1$	$y = 2$	$P_X(x)$
$x = 1$	$\frac{4}{5}$	$\frac{3}{5}$	$\frac{7}{10}$
$x = 2$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{10}$

$$E[X|Y = 1] = 1.\frac{4}{5} + 2.\frac{1}{5} = \frac{6}{5}$$

$$E[X|Y = 2] = 1.\frac{3}{5} + 2.\frac{2}{5} = \frac{7}{5}$$

Law of Total Expectation

$$E[X] = \sum_y E[X|Y = y]P_Y(y)$$

$$E[Y] = \sum_x E[Y|X = x]P_X(x)$$

Example 6.17. Sam will read either one chapter of his probability book or one chapter of his history book. If the number misprints in a chapter of the probability book is Poisson distributed with mean 2 and if the number of misprints in the history chapter is Poisson distributed with mean 5. Then assuming Sam is equally likely to choose either book, what is the expected number of misprints he will come across?

Solution. Let X denote number of misprints, and

$$Y = \begin{cases} 1, & \text{if Sam chooses his history book} \\ 2, & \text{if Sam chooses his probability book} \end{cases}$$

Then,

$$\begin{aligned} E[X] &= E[X|Y = 1]P[Y = 1] + E[X|Y = 2]P[Y = 2] \\ &= 5 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} \\ &= \frac{7}{2}. \end{aligned}$$

Example 6.19. A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that leads him to the mine after five hours. Assuming that the miner is at all times equally likely to choose any one of these doors, what is the expected length of time until the miner reaches safety?

$$E[X] = \frac{1}{3}(E[X|Y = 1] + E[X|Y = 2] + E[X|Y = 3])$$

$$= \frac{1}{3}(2 + 3 + E[X] + 5 + E[X])$$

$$E[X] - \frac{2}{3}E[X] = \frac{10}{3}$$

$$E[X] = 10.$$

Example 6.18 (The mean and variance of Geometric Distribution). A coin having probability p of coming up head is to be successfully flipped until the first head appears. What is the expected number of flips required and the variance of number of flips?

$$P_X(x) = (1 - p)^{k-1}p \quad k \geq 1$$

$$E[X] = \sum_{k=1}^{\infty} k(1 - p)^{k-1}p, \text{ and } \quad Var[X] = \sum_{k=1}^{\infty} (k - E[X])^2(1 - p)^{k-1}p$$

$$Y = \begin{cases} 1, & \text{if the first flip results in a head} \\ 0, & \text{if the first flip results in a tail} \end{cases}$$

$$\begin{aligned} E[X] &= E[X|Y = 1]P[Y = 1] + E[X|Y = 0]P[Y = 0] \\ &= pE[X|Y = 1] + (1 - p)E[X|Y = 0] \end{aligned}$$

$$E[X|Y = 1] = 1$$

$$E[X|Y = 0] = E[1 + X] = 1 + E[X]$$

$$\begin{aligned} E[X] &= p \cdot 1 + (1 - p) \cdot (1 + E[X]) \\ &= p + 1 - p + E[X] - p \cdot E[X] \end{aligned}$$

$$\text{or, } p \cdot E[X] = 1$$

$$\text{or, } E[X] = \frac{1}{p}.$$

Bayes Theorem for Discrete Random Variables

If X and Y are two Random variables with

$$P_X(x)$$

$$P_{Y|X}(y|x)$$

Then applying the sum and product rules, the Bayes formula is

$$P_{X|Y}(x|y) = \frac{P_{Y|X}(y|x)P_X(x)}{\sum_x P_{Y|X}(y|x)P_X(x)}$$

